

DVP&R; (virtual) VBF-T2R1SINGLEMODEL-DVPR-01

Virtual Battery Factory — t2_r1_singlemodel — 2026-08-25

Design Verification Plan and Report (virtual) — VBF-T2R1SINGLEMODEL-DVPR-01

> Virtual-test version: all results are simulation outputs; uncovered conditions honestly
> "N/A (requires physical experiment)". Generation date 2026-08-25.

Item	Condition	Result value	Determination	Source
1C discharge capacity	1C CC to 2.5 V, 298.15 K, DFN	6.9555 Ah	informational (no capacity criterion in entry 0)	cell/r3_1c_dfn.json
5C discharge capacity	5C CC to 2.5 V, DFN	0.4939 Ah (7% of 1C; voltage collapses to cut-off in ~71 s — severely rate-limited; informational, no 5C criterion)	informational (no 5C criterion in entry 0)	cell/r3_5c_dfn.json
4C fast-charge temperature rise	4C CC to 4.2 V, 45 C start, lumped thermal	T_max 342.5 K (rise +24.3 K)	informational (T_max monitored, no red line)	cell/r3_4c45.json
4C fast-charge plating	same + plating module	ap_min +0.112 V (45 C); +0.111 V (25 C conservative start)	PASS (plated == false)	cell/r3_4c45.json / cell/r3_4c25_conservative.json
Voltage window	parameter set cut-offs	2.5 – 4.2 V	informational	parameter set
Low-temperature retention	1C discharge, 253.15 K (-20 C), DFN	99.59 % vs 298.15 K	PASS (>= 90 %)	cell/derived_retention_lowT_r3.json
SEI growth @100 cycles	aging 1C cycling, ec reaction limited, DFN	194.56 nm	PASS (<= 500 nm)	cell/r3_aging100.json
SEI growth @500 cycles	aging 1C cycling, 500 completed cycles, DFN	439.74 nm	PASS (<= 550 nm)	cell/r3_aging500_d1e19.json
Overcharge to thermal runaway	0.5C charge to 4.7 V (overcharge protocol)	DFN solver failure in 4.7 V charge phase (IDA, recorded verbatim); TR coupling on partial-run T_max 322.7 K -> not triggered	conditional (virtual test incomplete; physical test required)	cell/r3_overcharge.json / cell/r3_tr.json
Nail penetration	5 W short-circuit heat source, lumped TR ODE	triggered, T_max 408.4 K @ 1159 s	informational (safety screen)	cell/r3_tr_nail.json
Crush / drop / storage life	—	N/A (beyond pure simulation boundary, requires physical experiment)	N/A	—

Conclusion

All entry-0 criteria PASS (energy density, SEI100, SEI500, low-T retention, 4C plating).
Uncovered: overcharge-to-4.7 V virtual test (DFN solver failure recorded; physical overcharge test recommended), nail/crush/drop physical confirmation, long-term storage.